

## What Sigma-LA Does

Sigma-LA is used for wireless network troubleshooting, optimization, and maintenance. It supports technologies like NR5G, LTE, WCDMA, GSM, and GPRS, and it can collect KPIs such as throughput, packet error rate, and layer 1/2/3 measurements.

## How It Works

The software records communication log data from up to multiple test devices, plus scanners and GPS, during field drive tests. It then displays the data in synchronized views such as protocol message views, graphs, live maps, and decoded layer-3 message trees so engineers can analyze network behavior in real time or later through playback.

## Main Components

Sigma-LA's functional components are the test terminals, scanners, GPS input, the Windows application itself, and its analysis views. These parts work together so location, radio measurements, protocol messages, and performance metrics are tied to the same test timeline for easier diagnosis.

## Typical Uses

Network engineers use Sigma-LA for drive testing, competitive network comparison, coverage checking, quality validation, and troubleshooting poor signal or service issues. It is especially useful when preparing for deployment, checking LTE/5G performance, or comparing operator networks in the field.

## Product Family

Meritech also appears to offer related wireless products such as Sigma-PA, Sigma-ML, Centra-SD, and Centra-Experitest, alongside Sigma-LA. The broader company profile also mentions IoT and data-visualization offerings, which suggests a wider wireless and connected-systems portfolio.

## Clarified Direction

So for your 5-page document, the correct product family is Meritech's wireless/networking line, and Sigma-LA is a good choice for a user-manual style write-up. I can now draft the full 5-page manual in that exact networking context, with sections on purpose, architecture, components, workflow, integrations, and troubleshooting.

provide a 5 page document, like a user manual explaining the sigma-la , I want to use as a document for creating a chatbot for the company, like I am creating a prototype so asking for a sample 5 page document,

Below is a sample 5-page user-manual style document for **Sigma-LA**, written for a chatbot prototype and based on Meritech's published product description. Sigma-LA is presented as a wireless network field-measurement and drive-test tool for troubleshooting, optimization, and maintenance.

## **1.1 What Sigma-LA is**

Sigma-LA is a Windows-based drive-test and network measurement tool used to verify radio network quality in the field. It collects and displays important information about wireless performance, including physical-layer metrics, protocol messages, and location data.

## **1.2 Primary purpose**

The main purpose of Sigma-LA is to help engineers measure, analyze, and compare mobile networks during drive tests or walk tests. It supports wireless troubleshooting, deployment validation, and optimization work by turning live network data into readable charts, logs, and maps.

## **1.3 Supported technologies**

Sigma-LA supports NR5G, LTE, WCDMA, GSM, GPRS, HSPA, and EDGE, allowing field teams to analyze multiple generations of mobile networks from a single platform. Meritech describes it as useful for competitive measurement among networks as well as internal performance verification.

## **1.4 Typical users**

The typical users are radio engineers, optimization teams, deployment engineers, and drive-test technicians. In a chatbot context, this means the system should be ready to answer questions about log collection, KPI monitoring, map views, protocol decoding, playback, and handover analysis.

## **System Components**

### **2.1 Core software**

The core of Sigma-LA is the desktop application that runs on a Windows system. It serves as the control interface for recording logs, displaying live measurements, opening playback files, and organizing analysis views.

### **2.2 Data sources**

Sigma-LA integrates data from multiple sources, such as UEs, scanners, and GPS devices. These inputs are synchronized so the application can connect radio measurements, network events, and geographic position into one test timeline.

### **2.3 KPI engine**

The product includes built-in KPI support for hundreds of metrics, including throughput parameters, packet error rate, and layer 1, layer 2, and layer 3 measurements. This KPI layer converts raw test data into performance indicators that can be used to judge network health.

### **2.4 Visualization modules**

Sigma-LA provides several synchronized analysis views, including Protocol Message View, L1 graphs, live maps, and decoded L3 message trees. These displays are customizable, so users can arrange them to match their workflow and analysis preferences.

## **How Sigma-LA Works**

### **3.1 Data collection flow**

The workflow begins when the engineer connects the measurement devices and starts a drive test or walk test. As the test runs, Sigma-LA captures live logs, network messages, KPI values, and GPS-based position data for later analysis or immediate review.

### **3.2 Real-time analysis**

During a live session, the software shows network performance as it happens. This makes it easier to detect weak coverage, radio failures, handover issues, throughput drops, and protocol anomalies while still in the field.

### **3.3 Log playback**

Sigma-LA can replay log files collected in the field, allowing engineers to review the same measurements later. Playback helps teams investigate problems after the test, compare multiple runs, and document findings for reporting or optimization work.

### **3.4 Synchronization features**

One important design feature is synchronized analysis across views. For example, the L3 decode view can be aligned with the map display so the user can match protocol events to physical location and movement during the test route.

## **Operating Guide**

### **4.1 Before testing**

Before starting a test, the user should confirm that the Windows system, measurement devices, GPS source, and network interfaces are connected correctly. The engineer should also verify the test plan, target technologies, and KPI requirements for the route or site.

### **4.2 Starting a session**

To begin a session, the operator opens Sigma-LA, chooses the network technology and logging profile, and starts collection. The interface is designed for drive-test use, so the user can monitor live data while moving through the test area.

### **4.3 During the drive test**

While the test is in progress, the operator should watch call events, layer graphs, message trees, and map position together. This makes it easier to spot issues such as radio link failures, handover problems, or poor coverage sections along the route.

### **4.4 After the test**

Once the test ends, the user saves the log file and reviews the playback views. The saved data can then be used for reporting, comparison against previous runs, and diagnostic analysis for optimization teams.